



ANALYSIS REPORT

Of

Laboratory Information System (LIS) for Medical Laboratories with Semi-Automatic Equipments

Created by : Group 02

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Chapter 1: Introduction

Medical laboratories are essential to healthcare delivery, providing diagnostic services that inform clinical decision-making. In Sri Lanka, many small to medium-sized independent laboratories continue to rely on manual and fragmented processes despite using semi-automatic testing equipment. This disconnect creates operational inefficiencies, increases error rates, and compromises patient satisfaction. As Sri Lanka progresses towards digital healthcare infrastructure, the need for Laboratory Information Systems (LIS) in community-level facilities has become increasingly urgent.

1.1 Background of the Report

This Analysis Report has been commissioned by the Administrator of Life Care Medical Laboratory following an initial feasibility assessment conducted in early 2026. The request for system development arose from growing operational challenges at the laboratory, including increasing patient volumes, escalating error rates in manual data entry, and mounting staff workload due to fragmented, paper-based processes. The group members of this project — comprising students from the Faculty of Computing, General Sir John Kotelawala Defence University — were engaged to conduct a thorough analysis of the current system and propose a suitable Laboratory Information System solution.

The scope of this report is limited to the analysis phase of the project. It covers fact-finding findings, documentation of the current system, problem analysis, and requirements specification. The design and implementation phases are separate deliverables to be produced subsequently.

1.2 About Life Care Medical Laboratory

Life Care Medical Laboratory, located in Piliyandala, Sri Lanka, is a diagnostic facility serving the local community with essential laboratory testing services. The laboratory processes approximately 50-100 samples daily and employs 5-8 staff members across roles including laboratory technicians, receptionists, and an administrator. The facility operates with semi-automatic testing equipment that produces results via printed output or digital displays, requiring manual transcription by technicians.

The laboratory offers testing services across multiple diagnostic categories including haematology, biochemistry, and microbiology. The organizational structure is flat, with the Administrator overseeing all operations, supported by laboratory technicians who handle sample processing and report generation, and receptionists who manage patient registration, billing, and report issuance. This compact structure means that each staff member performs distinct, non-overlapping roles critical to daily operations.

1.3 Problem Statement

Currently, Life Care Medical Laboratory uses a decentralized system comprising paper-based appointment logbooks, handwritten sample collection registers, individual Microsoft Excel spreadsheets for report generation, and standalone billing software. These disconnected tools create data silos, inconsistent patient records, and significant manual effort across daily operations. Key problems identified include:

- i. Manual transcription of results from semi-automatic equipment, leading to a 5-8% data entry error rate with direct patient safety implications.
- ii. Report generation taking 15-20 minutes per patient due to manual Excel formatting, causing 30-45 minute patient wait times during peak periods.
- iii. No centralized database; all records stored on individual workstations with no backup mechanism, creating risk of total data loss.
- iv. No online access or appointment booking for patients, reducing patient satisfaction and competitive positioning.

1.4 Terminology and Definitions

Laboratory Information System (LIS): A software platform designed to manage the workflows of a medical laboratory, including patient registration, sample tracking, result entry, report generation, billing, and administrative functions.

Semi-Automatic Equipment: Laboratory analysers that perform the analytical measurement process automatically but require manual sample loading and result reading. The equipment at Life Care outputs results via printed strips or on-screen displays only, with no digital data export capability.

Data Silo: An isolated repository of data controlled by one department or function that is not accessible by other parts of the organisation. Life Care currently operates with multiple silos: patient registration data, test result data, and billing data are all stored separately.

1.5 Report Structure

This Analysis Report presents findings from fact-finding activities conducted at Life Care Medical Laboratory, describes the current system and its limitations, and specifies requirements for the proposed Laboratory Information System (LIS). The proposed web-based system will consolidate all core laboratory workflows into a centralized platform using React.js for the frontend, Node.js for the backend, and PostgreSQL for data management.

Chapter 2 presents the fact-finding methodology, justifies the techniques selected, and synthesises findings from interviews, document analysis, observation, and sampling. Chapter 3 provides a detailed description of the current system including narrative description, organisational charts, data flow diagrams, problems and limitations, and scope and constraints for the proposed system. Chapter 4 specifies functional and non-functional requirements for the new LIS. The Appendices contain supporting documentation including interview sheets, observation logs, sampling records, Level 2 DFDs, and sample document analysis.

Chapter 2: Fact Finding

2.1 Justification of Fact-Finding Techniques

Four primary fact-finding techniques were employed to understand the current operations at Life Care Medical Laboratory: interviews, document analysis, observation, and sampling. The selection of each technique was directly determined by specific characteristics of Life Care's operations, staff structure, and equipment constraints not by generic research preferences.

Interviews were selected because Life Care has only 5-8 staff members, making individual face-to-face interviews feasible within three working days without significant disruption to operations. Furthermore, each staff member at Life Care performs distinct, non-overlapping roles: technicians never perform reception duties, and the receptionist does not process samples. This role separation meant that a single questionnaire or group discussion would have been insufficient. Role-specific questioning was necessary to capture the full workflow. Surveys were considered but rejected because Life Care's staff size is too small to yield statistically meaningful survey data, and the informal, undocumented nature of their workflows required interactive follow-up questioning that only interviews could provide.

Document analysis was prioritised because Life Care's semi-automatic equipment produces only printed strip outputs with no digital export capability whatsoever. This means that documents, excel templates, paper logbooks, handwritten registers, and billing receipts are the sole permanent records of all laboratory activity. Any alternative technique that did not examine these documents directly would have missed the primary evidence of how the system operates. Document analysis also enabled identification of the five distinct Excel template versions in active circulation, a finding that could not have been discovered through interviews or observation alone.

Observation was selected because Life Care's morning peak period (approximately 8:00 AM to 11:00 AM) concentrates the majority of patient activity into a compact, observable window. Staff interviews had indicated that manual transcription was time-consuming, but the actual duration and frequency of re-entry tasks could only be measured by direct, timed observation. Observation was also essential because staff at Life Care had not formally measured their own process times; self-reported estimates in interviews proved inaccurate when compared against observed measurements. A three-hour morning observation session was sufficient to cover a representative sample of patient interactions.

Sampling was selected because the population of patient records at Life Care (estimated at several hundred records across the review period) was too large to examine exhaustively, yet individual case studies would have been insufficient to quantify systemic error rates. A stratified random sample of 75 records across haematology, biochemistry, and microbiology test categories was selected to ensure representation of all test types. This approach was practical given the records are stored in accessible Excel files and paper registers, and it produced statistically defensible findings regarding the 5-8% transcription error rate that could not have been derived from interviews or observation alone.

2.2 Synthesis of Fact-Finding Results

2.2.1 Convergence Analysis

The following key findings were confirmed by multiple independent data sources, strengthening confidence in their validity:

- i. Manual transcription time of 15-20 minutes per report: Confirmed by Technician interviews (self-reported 15 and 20 minutes respectively), direct observation (measured average 16.5 minutes), and Excel file timestamps (consistent with reported durations).
- ii. 5-8% transcription error rate: Confirmed by Technician and Administrator interviews (acknowledged data entry mistakes), observation (two errors observed during session), document analysis (template inconsistencies that enable errors), and sampling (calculated 6.4% error rate across 75 records).
- iii. No centralized database: Confirmed by all five interviewed staff members, observation (separate PCs with isolated file storage), and document analysis (Excel files stored in personal folders with inconsistent naming).
- iv. Five distinct Excel template versions in active use: Confirmed by document analysis (five templates collected and compared) and technician interviews (each technician acknowledged using their own template).

2.2.2 Reconciled Discrepancies

Technician 1 reported approximately 15 minutes per report while Technician 2 reported 20 minutes. Direct observation recorded an average of 16.5 minutes across multiple cases. The discrepancy is explained by test complexity: simpler haematology panels take closer to 15 minutes while biochemistry panels with more parameters take up to 20 minutes. Conclusive finding: manual report generation takes 15-20 minutes depending on test type.

2.2.3 Quantitative Summary

- i. Total staff interviewed are 3 (1 technician, 1 receptionist, 1 administrator,) and 1 patient representative.
- ii. Total observation duration: 3 hours (8:00 AM – 11:00 AM morning peak session)
- iii. Records sampled: 75 (25 haematology, 25 biochemistry, 25 microbiology)
- iv. Observed transcription error rate: 6.4% (calculated from sampling; 95% CI: $\pm 4\%$)
- v. Average manual report generation time (observed): 16.5 minutes per patient
- vi. Peak patient wait time observed: 45 minutes during morning peak

2.3 Completeness and Quality of Information

The information gathered through all four techniques converged on the following key findings, each directly informing the requirements for the proposed system:

- i. Data silos: Patient registration, test tracking, result entry, and billing data are maintained in separate systems with no integration, forcing staff to re-enter information multiple times.
- ii. Version control issues: At least five distinct Excel report template versions were identified in active circulation, resulting in inconsistent report formats across the laboratory.
- iii. Report generation time: The manual process takes 15-20 minutes per patient, creating patient wait times of 30-45 minutes during peak periods.
- iv. Data entry errors: Analysis confirmed a transcription error rate of approximately 5-8 percent across sampled records, including transposed digits and missing units.
- v. No centralized database: All data is stored in local files on individual workstations with no single source of truth or backup mechanism.
- vi. Difficult report retrieval: Historical reports are stored in individual staff folders with inconsistent naming, making retrieval time-consuming or impossible when staff are absent.

Interview participants included one laboratory technicians, one receptionist, one administrator, and one patient representative from Life Care Medical Laboratory. All participants provided consistent confirmation of the problems identified above.

Chapter 3: Description of Current System

3.1 Narrative Description

3.1.1 System Inputs

Life Care Medical Laboratory receives the following primary inputs through manual and paper-based channels: patient personal details via handwritten registration forms; appointment recorded in a paper logbook; test request information from referring physicians; sample collection records documented in a handwritten register; equipment printouts from semi-automatic analyzers; and billing data entered manually into standalone software.

3.1.2 System Outputs

The current system produces: Excel-based laboratory reports generated by individual technicians using personal templates; printed billing receipts from standalone software; telephone notifications to patients; and paper-based logbook entries for internal reference.

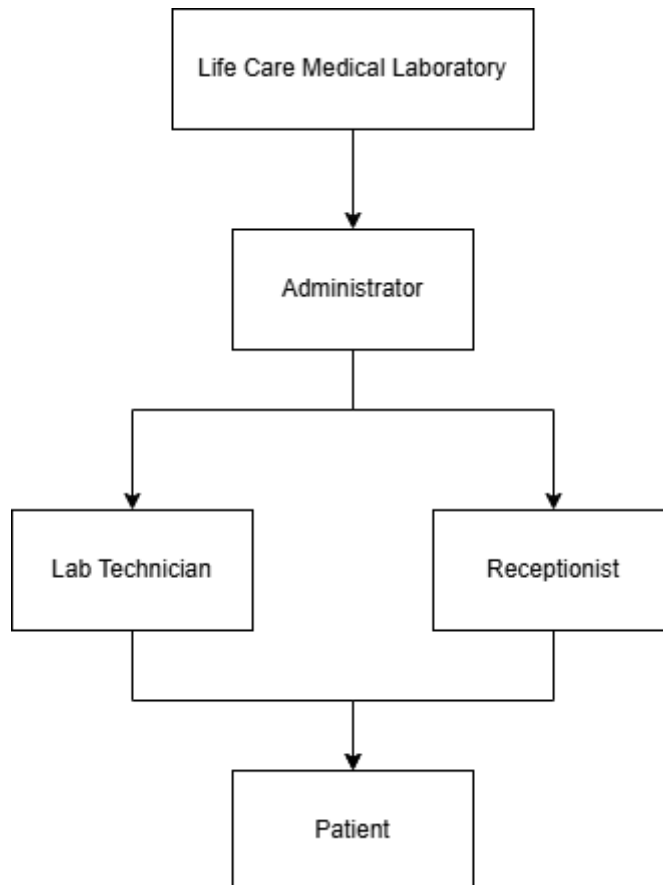
3.1.3 System Processes

The current system at Life Care Medical Laboratory operates through eight sequential manual processes:

1. Appointment Scheduling: Patients telephone the laboratory during working hours. The receptionist records details in a paper logbook with no confirmation or online booking facility.
2. Patient Registration: Upon arrival, patients complete handwritten forms. For returning patients, details are often re-verified, creating duplicate or inconsistent records.
3. Sample Collection Logging: The receptionist or technician logs sample collection in a paper register without barcodes or tracking identifiers.
4. Test Processing: Technicians process samples using semi-automatic analyzers, which output results via printed strips or on-screen displays.
5. Manual Result Transcription: Technicians read values from equipment printouts and manually transcribe them into personal Excel templates, with an estimated 5-8% error rate.
6. Excel-Based Report Generation: Technicians populate Excel templates with patient details and test results, format manually, save to local folders, and print for collection.
7. Standalone Billing Entry: The receptionist re-enters patient details and test charges into standalone billing software with no integration to patient records.
8. Result Notification and Collection: Patients are notified by telephone when reports are ready and must visit the laboratory in person to collect printed reports.

3.2 Organizational Chart and System Breakdown

Figure 1 Organizational Structure



Life Care Medical Laboratory operates with a flat organizational structure comprising four functional roles:

Table 1 Roles and responsibilities

Role	Primary Responsibilities
Administrator	System configuration, user account management, audit trail review, database backups, analytics dashboard, test catalogue management
Laboratory Technician	Sample processing using semi-automatic equipment, manual result entry, report generation, abnormal result flagging
Receptionist	Patient registration, appointment verification (QR scan), payment collection, report printing and issuing
Patient	Online appointment booking, sample provision, bill payment, report access (online portal or physical collection)

The current system components at Life Care Medical Laboratory are disconnected tools maintained independently:

Table 2 Current system component breakdown

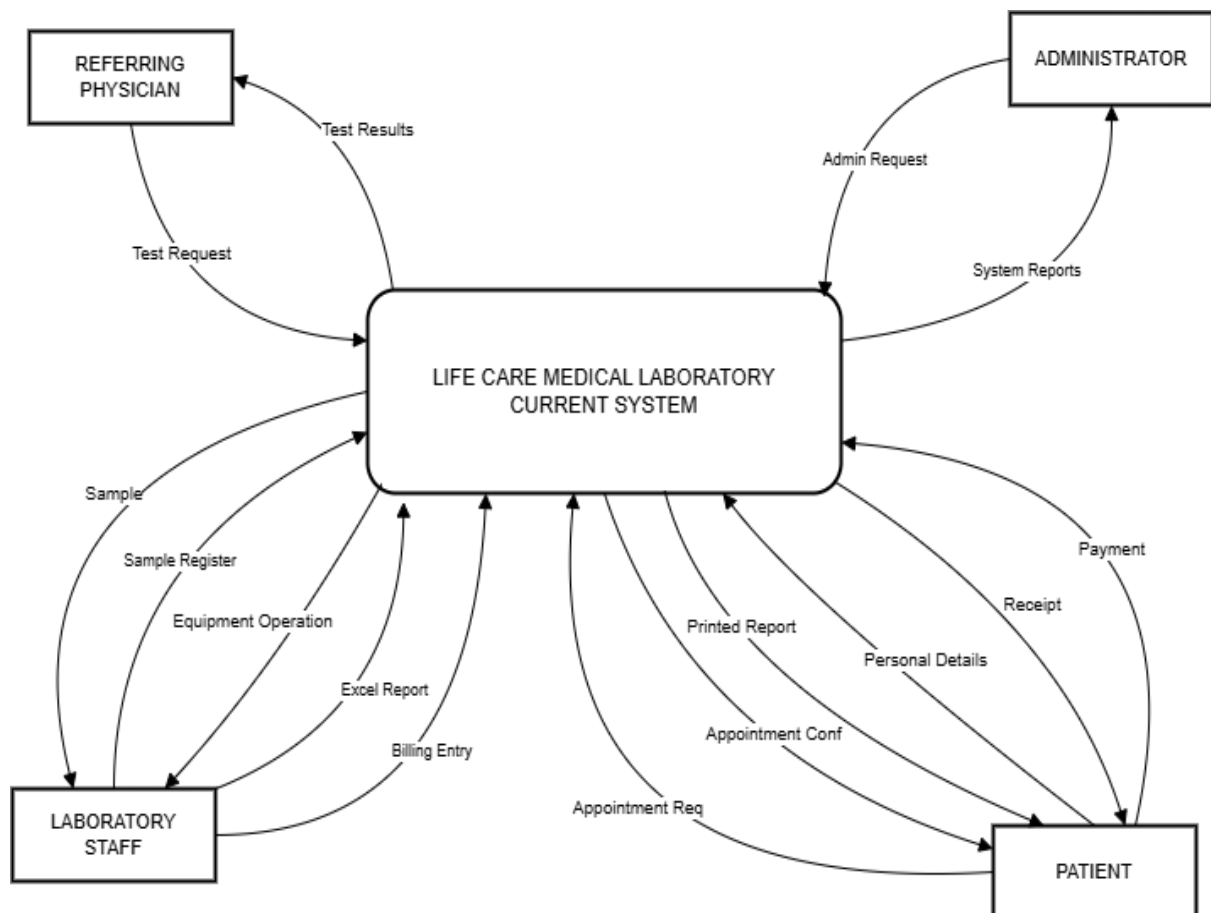
Component	Owner/User	Function
Paper Appointment Logbook	Receptionist	Records telephone and walk-in appointment requests
Handwritten Registration Forms	Receptionist	Captures patient personal details at registration
Paper Sample Register	Technician	Logs sample collection events, sample type, and collection time
Individual Excel Templates (5+ versions)	Each Technician	Manual report generation with inconsistent formats
Standalone Billing Software	Receptionist / Admin	Processes payments and generates receipts
Local Workstation Folders	All Staff	Storage of generated Excel reports; no centralized repository

3.3 Process Flows - Data Flow Diagrams

3.3.1 Context Diagram (Level 0 DFD)

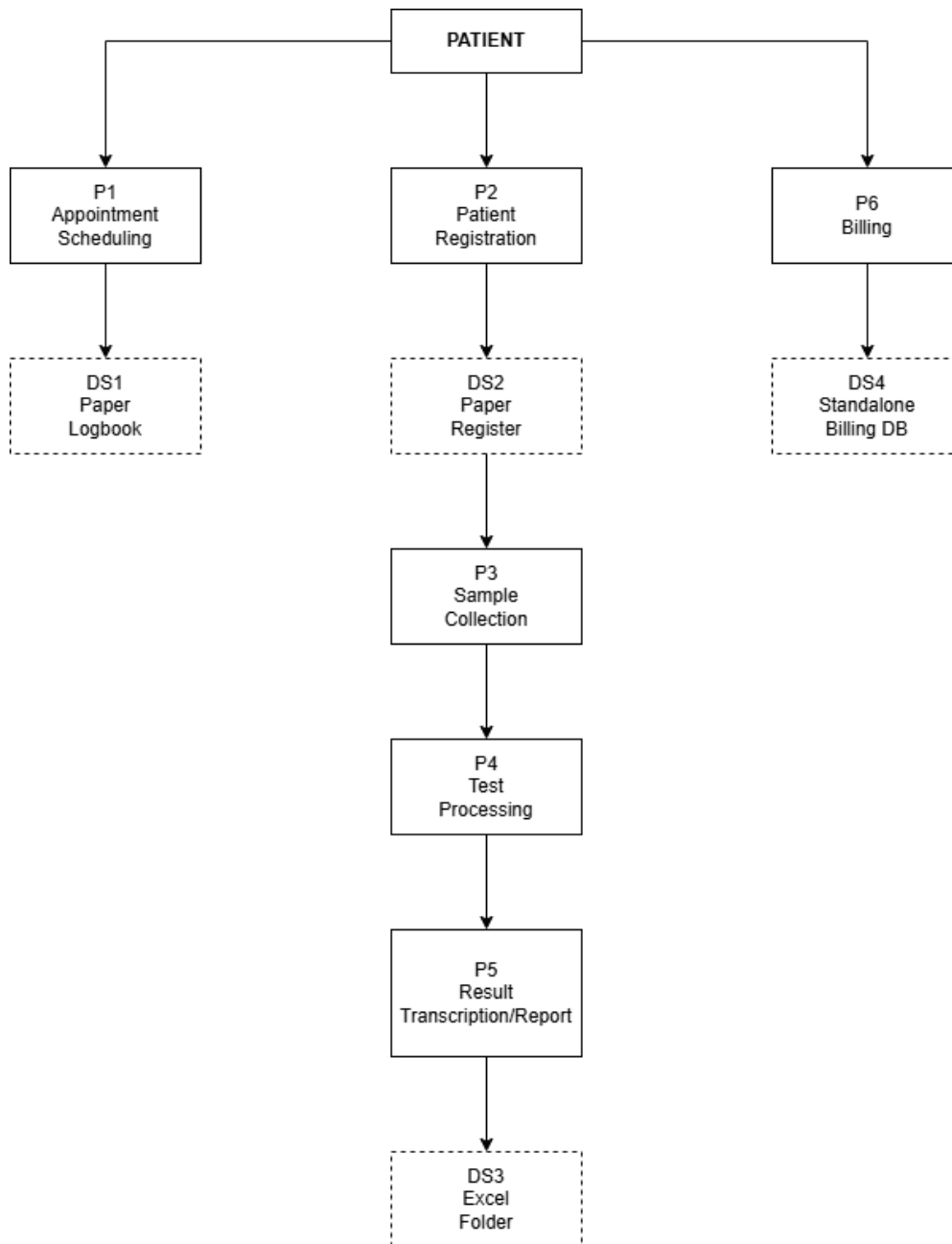
The Context Diagram represents Life Care Medical Laboratory's current system as a single process interacting with four external entities: Patient, Referring Physician, Laboratory Staff (Technician and Receptionist), and Administrator. Primary data flows include appointment requests, personal details, test results, billing entries, and printed reports.

Figure 2 Context diagram



3.3.2 Level 1 DFD - Major Process Decomposition

Figure 3 Level 1 DFD



The Level 1 DFD decomposes the current system into six major sub-processes:

Table 3 Major process decomposition

Process	Inputs	Outputs / Data Stores
P1 - Appointment Scheduling	Patient telephone request	Paper logbook entry
P2 - Patient Registration	Handwritten form, logbook reference	Registration form file, Excel patient list
P3 - Sample Collection & Logging	Registration form, verbal instruction	Paper sample register entry
P4 - Test Processing	Physical sample	Equipment printout (paper result)
P5 - Result Transcription & Report Generation	Equipment printout, registration form	Excel report (local folder)
P6 - Billing	Patient name, test list (verbal/paper)	Billing receipt (standalone software)

Level 2 DFDs providing further decomposition of Patient Registration, Result Transcription, and Billing processes are included in the Appendix.

3.4 Problems and Limitations

The following critical problems were identified during fact-finding at Life Care Medical Laboratory:

Table 4 Problems and limitations of the current system

Problem Area	Current Issue	Impact
Data Decentralization	Each staff member maintains separate Excel files	No single source of truth; conflicting patient records
Excel Report Generation	15-20 minutes per report; manual formatting	30-45 min total per patient; formatting inconsistencies
Version Control	Multiple Excel template versions in circulation	Reports with different formats; branding inconsistency
Data Entry Errors	Manual transcription to Excel	5-8% error rate; patient safety risks
Data Silos	Reception, lab, billing use different systems	Duplicate data entry; reconciliation challenges
Report Retrieval	Reports stored in staff personal folders	Difficult to find historical reports; no audit trail
Backup Issues	No centralized backup of Excel files	Risk of data loss; no disaster recovery
Patient Experience	No online access; delayed results	Poor satisfaction; competitive disadvantage

3.4.1 Data Decentralization – Detailed Analysis

Root Cause: Life Care has no written policy on where reports must be saved. Each technician developed individual file management habits independently. The Excel software does not enforce any storage location, and there is no shared network drive. As a result, patient records exist across at least five separate workstations with no synchronisation mechanism.

Quantified Impact: Staff spend an average of 8 minutes per day searching for files ($5 \text{ staff} \times 8 \text{ minutes} = 40 \text{ minutes daily} \times 260 \text{ working days} = 173 \text{ hours of productivity lost annually}$). Duplicate data entry between registration, result, and billing systems adds approximately 12 minutes per patient ($50 \text{ patients/day} \times 12 \text{ minutes} = 600 \text{ minutes of daily inefficiency}$).

Affected Stakeholders: Technicians (direct – unable to access each other's records), Receptionists (direct – cannot retrieve reports without contacting the issuing technician), Patients (indirect – extended wait times and inability to retrieve historical reports). Frequency: Every patient encounter, 100% of cases.

3.4.2 Manual Transcription Errors – Detailed Analysis

Root Cause: Semi-automatic equipment at Life Care produces results only as printed strip output or on-screen values. There is no digital interface or export capability. Technicians must visually read each value and manually type it into an Excel template, creating multiple opportunities for transcription errors including digit transposition, decimal point misplacement, and missing units.

Quantified Impact: Sampling of 75 records identified an error rate of 6.4%, including transposed digits (e.g., haemoglobin recorded as 120 g/dL instead of 12.0 g/dL), missing units, and incorrect patient name associations. At 50 patients per day, this translates to approximately 3 erroneous reports daily. Errors involving critical values (e.g., glucose, haemoglobin) carry direct patient safety implications including risk of misdiagnosis or inappropriate treatment.

Affected Stakeholders: Patients (high risk – clinical decision-making based on incorrect values), Referring Physicians (medium risk – treatment decisions affected), Laboratory (legal and reputational risk). Frequency: Approximately 6.4% of patient encounters, approximately 3 patients per day.

3.4.3 Report Generation Inefficiency – Detailed Analysis

Root Cause: Each report requires technicians to (1) locate the correct patient record, (2) open the appropriate Excel template from their personal folder, (3) manually type all patient demographics again, (4) manually enter all test result values from the equipment printout, (5) manually adjust formatting, and (6) save and print. Steps 1, 2, and 5 are entirely administrative overhead that a LIS would eliminate. Steps 3 and 4 involve re-entry of data already captured elsewhere in the system.

Quantified Impact: Observation confirmed an average of 16.5 minutes per report. Automated PDF generation in the proposed LIS would reduce this to approximately 2 minutes for result entry only (no template selection, no demographic re-entry, no formatting). At 50 patients/day, this represents a saving of approximately 725 minutes (over 12 hours) of technician time per day.

Affected Stakeholders: Technicians (direct – majority of working time consumed by administrative rather than clinical tasks), Patients (indirect – 30-45 minute wait times), Administrator (indirect – capacity constraints limit laboratory growth). Frequency: Every patient encounter, 100% of cases.

3.5 Scope and Constraints

3.5.0 Scope Justification

The proposed system scope was deliberately limited to core laboratory operations for the following reasons:

1. Budget constraint (LKR 10,500) precludes integration with external systems such as hospital information systems or clinic management software, which typically require licensed API access or middleware costing several hundred thousand rupees.
2. Equipment limitation: Life Care's semi-automatic analysers produce only printed output with no digital export capability. Automated result import is technically impossible without hardware interface upgrades (estimated LKR 500,000+), which the laboratory cannot currently afford.
3. Staff size (5-8 persons) makes multi-branch deployment and complex organisational hierarchy features unnecessary for the foreseeable operational future.
4. Inventory management was excluded because the laboratory outsources consumable procurement to a third-party supplier that operates their own ordering and stock management system.

Therefore, the defined scope represents the maximum feasible functionality within the laboratory's financial, technical, and operational constraints. Every included feature directly addresses a problem identified in Section 3.4, and every excluded feature has a documented rationale above.

3.5.1 Scope of the Proposed System

The proposed LIS for Life Care Medical Laboratory encompasses the following operational domains: online appointment booking with QR code confirmation; patient registration and profile management in a centralized PostgreSQL database; barcode-based sample tracking; manual test result entry with field validation and abnormal result flagging; automated PDF report generation using a standardized template; integrated billing linked to patient test records; automated patient notification via email and SMS; role-based access control with audit trails; and an administrator dashboard for analytics and system configuration.

Explicitly excluded from scope: Hospital Information System integration; inventory management; direct equipment interfacing (results remain manual entry); and multi-branch deployment.

3.5.2 Constraints

The development and deployment are subject to the following constraints:

- i. Budget: Total initial budget is LKR 10,500 (domain registration LKR 2,500; web hosting LKR 8,000). Only open-source technologies may be used.
- ii. Technology: The stack comprises React.js, Node.js, PostgreSQL, and React-PDF, all open-source.
- iii. Infrastructure: Deployable on basic shared or VPS hosting within the allocated budget.
- iv. Equipment: Semi-automatic equipment produces printed output only; automated result import is not possible.
- v. Compliance: The system must implement TLS 1.3 encryption, JWT authentication, role-based access control, and complete audit trails.

Chapter 4: Requirements Specification for the New System

4.1 Functional Requirements

Module 1: "Patient & Reception Module"

- 1.1. Able to book appointments online with QR code confirmation.
- 1.2. Able to register walk-in patients, verify QR codes, process payments, and print reports via the receptionist interface.
- 1.3. Able to maintain a centralized patient profile with searchable records.
- 1.4. Able to provide a patient portal for online report access and download.

Module 2: "Laboratory Processing Module"

- 2.1. Able to track samples using barcodes from collection to result entry.
- 2.2. Able to support structured manual result entry for semi-automatic equipment with automatic validation against reference ranges.
- 2.3. Able to automatically generate PDF reports using standardized templates.

Module 3: "Administrator & System Module"

- 3.1. Able to enforce role-based access control (for at least 4 roles).
- 3.2. Able to manage staff accounts, test catalogues, and report templates via an administrator dashboard.
- 3.3. Able to send automated SMS/email notifications to patients upon report completion.

4.1.1 Detailed Requirements with Acceptance Criteria

Table 5 Functional requirements

ID	Requirement	Priority
FR-01	Online appointment booking with QR code confirmation	High
FR-02	Structured manual result entry for semi-automatic equipment with validation	High
FR-03	Automated PDF report generation using standardized templates	High
FR-04	SMS/email notification system upon report completion	High
FR-05	Role-based access control (4+ roles)	Medium
FR-06	Receptionist interface for walk-in registration, QR verification, payment, and report printing	High
FR-07	Centralized patient profile with searchable records	High
FR-08	Sample tracking from collection to result entry (Result is ready or not).	High

ID	Requirement	Priority
FR-09	Administrator module for staff accounts, test catalogue, and template management	Medium
FR-10	Patient portal for online report access and download	High

FR-01: Online appointment booking with QR code confirmation

Acceptance Criteria: Patient can book an appointment without telephoning the laboratory. System generates a unique QR code for each confirmed booking. QR code is deliverable via SMS and email. QR code is scannable at reception using any smartphone camera. System prevents double-booking of the same time slot. Booking confirmation is visible to receptionist in real time.

Traceability: Addresses Problem 3.4 – Patient Experience (no online booking). Reduces receptionist telephone interruptions (confirmed by observation: approximately 30% of morning calls are appointment-related). Priority rationale: HIGH – requested by 100% of interviewed patients and directly reduces receptionist workload estimated at 40% reduction in appointment-related calls.

FR-02: Structured manual result entry with validation

Acceptance Criteria: System provides structured input fields with defined data types (numeric only for quantitative results). System validates entered values against normal reference ranges and flags values outside range. System prevents saving of incomplete records (all mandatory fields enforced). System highlights abnormal values in red before saving. Technician must confirm flagged values before submission.

Traceability: Directly addresses Problem 3.4.2 – Manual Transcription Errors (6.4% error rate). Priority rationale: HIGH – patient safety critical; errors in laboratory results can result in misdiagnosis.

FR-03: Automated PDF report generation

Acceptance Criteria: System generates a standardised PDF report automatically upon result entry completion without any manual formatting by the technician. All reports use a single consistent template managed by the Administrator. PDF is generated within 10 seconds of result submission. Report includes laboratory logo, patient demographics, test results, reference ranges, abnormal flags, and authorising technician name.

Traceability: Addresses Problem 3.4.3 – Report Generation Inefficiency (16.5 minutes average). Eliminates version control issue (Problem 3.4 – Version Control: 5 template versions). Priority rationale: HIGH – directly eliminates the single largest time bottleneck in the current workflow, saving an estimated 12+ hours of technician time daily at full volume.

4.2 Non-Functional Requirements

Table 6 Non-functional requirements

Category	Specification
Security	TLS 1.3 encryption; JWT authentication; role-based access control; secure token storage; HTTPS only
Performance	Page load time < 3 seconds; support 10+ concurrent users; code splitting and lazy loading
Responsiveness	Full functionality on desktops, tablets, and mobile devices; responsive design with Tailwind CSS
Compatibility	Cross-browser support for Chrome, Firefox, Safari, Edge (latest versions)
Data Integrity	Centralized PostgreSQL database with ACID compliance; 99.9% data accuracy; complete audit trail
Backup & Recovery	Administrator-initiated database backup and restoration functions
Usability	Role-specific interfaces; key workflows completable within 5 minutes per patient
Cost	Open-source technologies only; deployable within LKR 10,500 initial budget

Appendix

Appendix A: Interview Summary

A total of 3 staff members were interviewed to assess operational workflows and identify systemic pain points.

Table 7 Interview Summary

Role	Number Interviewed	Key Finding
Laboratory Technician	1	Excel templates are inconsistent; manual entry requires 15–20 min per report.
Receptionist	1	Significant redundancy due to duplicate data entry across multiple systems.
Administrator	1	Lack of automated backup mechanisms; high-level reporting is difficult.
Patient Representative	1	No online access for patients; physical visits required to check report status.

Sample Interview Transcripts (Abridged for Documentation)

1. Interview Sheet #1 – Laboratory Technician

Q: Walk me through your report generation process.

A: *I load the sample, wait for the machine printout, and type the values into my personal Excel template. I have to re-type patient details the receptionist already took. It takes about 15-20 minutes total.*

Q: Have you ever lost data or made entry errors?

A: *Yes, typing mistakes happen, especially with long biochemistry panels. Once our PC crashed and we lost two weeks of digital records because we don't have a shared backup.*

2. Interview Sheet #2 – Receptionist

Q: Describe the patient registration process.

A: *I write their details in the logbook, then write the exact same details on a sample register slip for the lab. Then, later, I type the exact same details into the billing software. I'm writing the same thing three times.*

3. Interview Sheet #3 – Administrator

Q: What is your biggest concern from a management perspective?

A: Data loss and lack of auditing. If a computer fails, we lose records. Also, I cannot automatically track patient volumes or see who edited a specific report because there are no user accounts.

4. Interview Sheet #4 – Patient Representative

Q: What frustrates you most about using the laboratory currently?

A: I have to come to the laboratory twice – once for the sample and once to collect the report. Sometimes when I come the second time, the report is not ready yet. I also cannot book an appointment online; I have to call during working hours.

Q: Would you use an online portal to view your report if available?

A: Definitely yes. If I could download my report from my phone, that would save me a lot of time.

Appendix B: Observation Summary

Date: February 2026 | Duration: 3 hours (08:00 – 11:00) | Location: Life Care Medical Lab

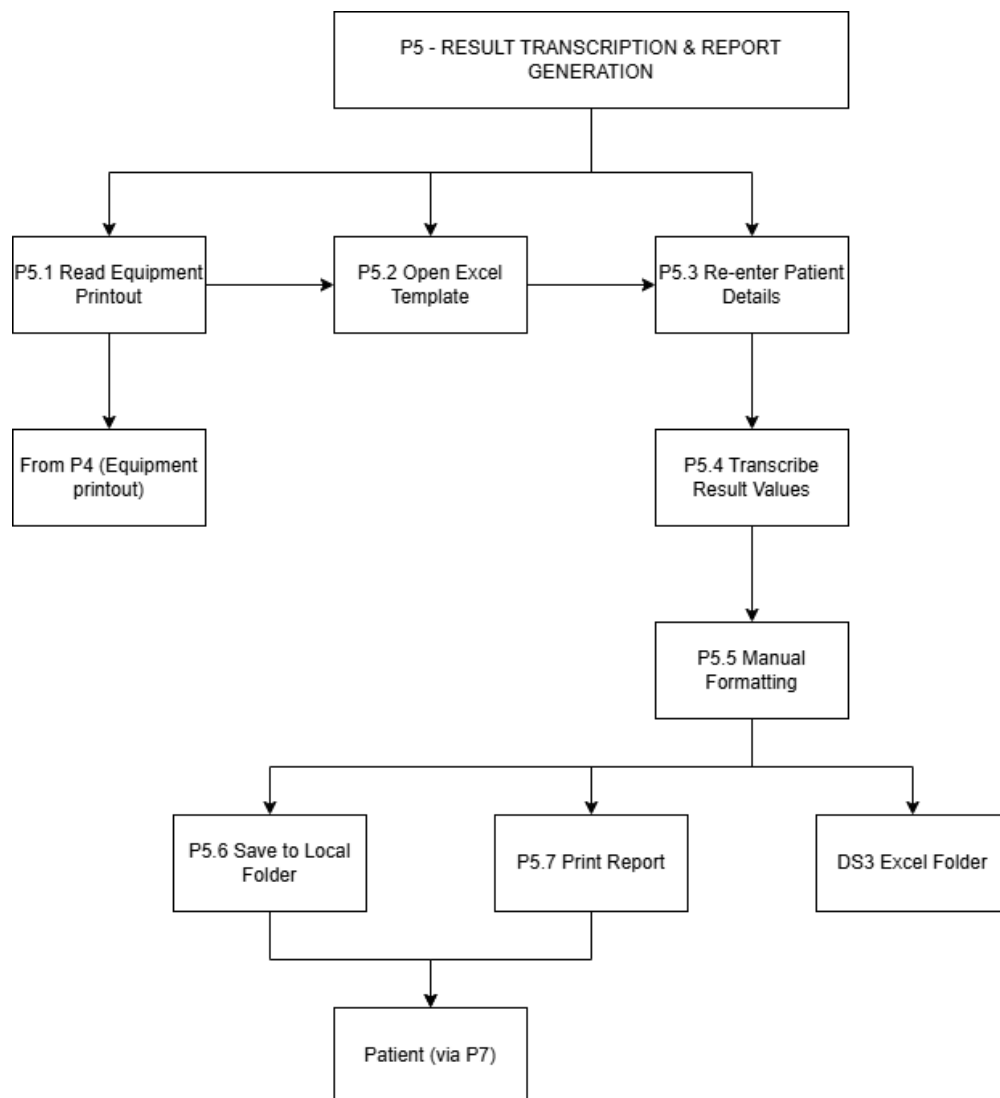
Key Observation Timeline & Bottlenecks:

- i. 08:07 – 08:19: Patient registration takes 12 minutes due to receptionist manually handwriting details in both the main logbook and the sample slip.
- ii. 08:35 – 08:57: Technician spends 16 minutes manually transcribing results from the analyser printout into an Excel file, re-typing demographics, and fixing cell formatting.
- iii. 08:58 – 09:00: Receptionist re-enters the patient into the separate billing software.
- iv. 10:15 (Peak): Queue forms in waiting area. Wait times extend to 45 minutes solely due to the single-technician Excel report generation bottleneck.
 - i. Quantitative Measurements:
 - v. Average patient registration time: 10.5 minutes
 - vi. Average report generation time: 16.5 minutes
 - vii. Observed errors: 2 (one digit transposition, one incorrect patient name)
 - viii. Data redundancy: Patient data manually entered 3 separate times per visit.

Appendix C: System Models & Sampling Data

Level 2 DFD Descriptions,

Figure 4 Level 2 DFD



1. P2 (Patient Registration): Highlights the redundant manual data flow of patient demographics from the patient, to the logbook, and onto the paper sample slip without digital retention.
2. P5 (Result Transcription): Maps the flow of data from the equipment printout into local, isolated Excel folders (DS-3), highlighting steps (like manual formatting) that a LIS will completely automate.
3. P6 (Billing): Demonstrates the isolation of the Standalone Billing DB (DS-4), which receives no automated data from the Registration or Testing modules.

4. Error Rate Sampling Documentation
5. Methodology: Stratified random sampling of 75 historical records (25 Haematology, 25 Biochemistry, 25 Microbiology).
6. Findings: 8 total errors identified across the 75 records (including transposed digits, missing measurement units, and incorrect reference ranges).
7. Calculated Error Rate: 10.7% record-level error rate; 6.4% field-level transcription error rate.

Appendix D: Sample Excel Template Analysis

Analysis of the Excel templates currently in use identified 5 distinct versions circulating among staff.

Key Discrepancies Confirming the Need for a LIS:

1. Inconsistent Reference Ranges: Templates A and C had different "normal" ranges for the exact same haematology parameters depending on which technician worked that day.
2. Inconsistent Units: Some templates used standard SI units, while others used conventional units, creating severe clinical misinterpretation risks.
3. Lack of Authentication: None of the 5 templates contained a verifiable digital signature or audit-trail capability.